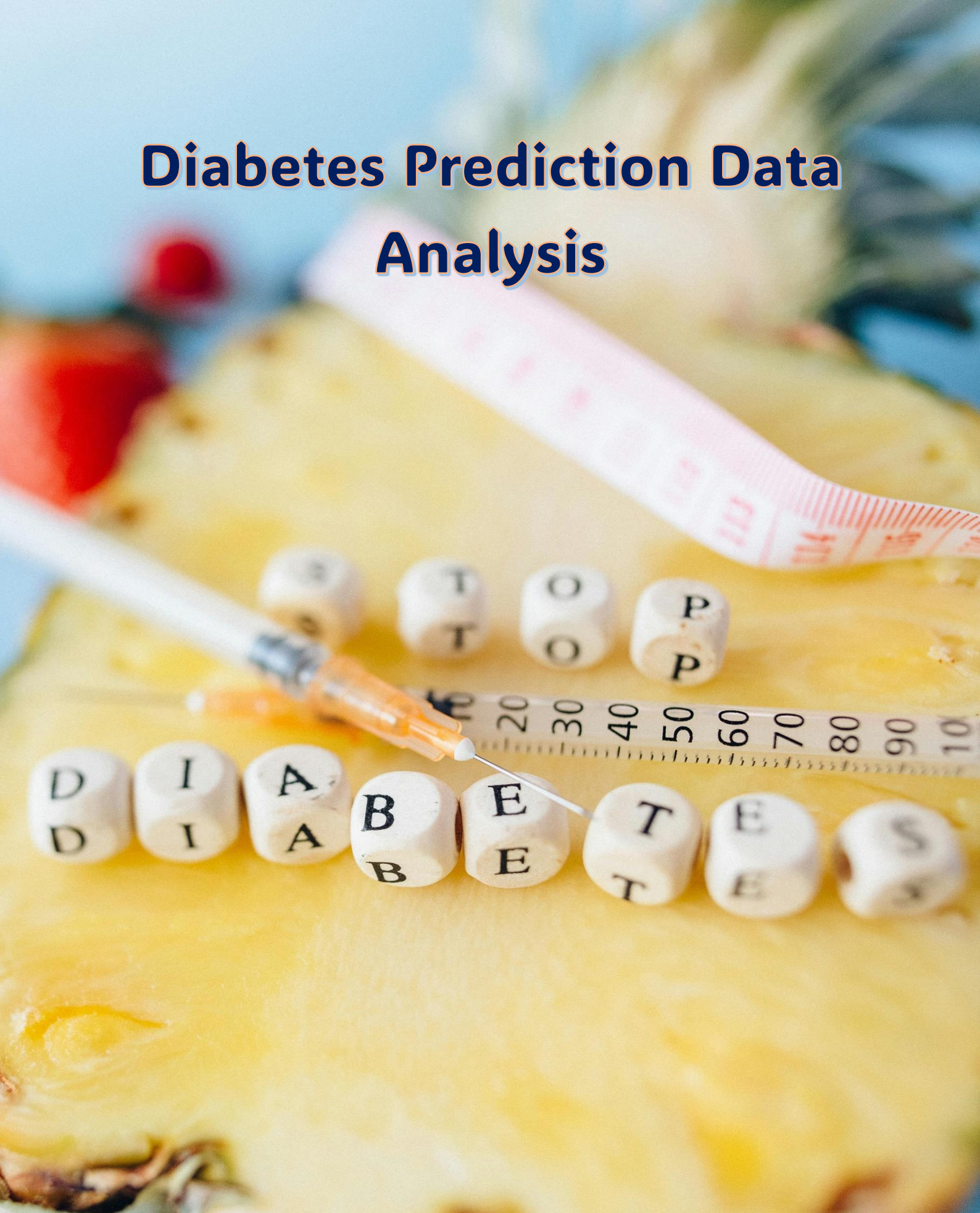


Diabetes Prediction Data Analysis



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Introduction

Diabetes is a common illness that affects many people and is a major public health issue both in the UK and throughout the world. It was estimated in 2017 that 3.7 million individuals in the UK had diabetes, which contributed to almost 700 deaths a week in the whole country [1]. As one of the main causes of disability and death in the UK, diabetes affects around 4.9 million individuals, according to the latest statistics.

With one in ten persons in the UK being diagnosed with the disease, the frequency of diabetes is concerning. It is estimated that by 2025, up to five million individuals in the UK may have diabetes if the present scenario continues [2]. Due to long-term tissue damage, diabetic patients are more likely to have medical complications such as kidney failure, eyesight loss, stroke, heart disease, and early death. Furthermore, diabetes has a significant financial impact; in the UK, the annual cost of treating diabetes is estimated to be £14 billion [3].

People who are at high risk for diabetes may be unaware of the risk factors. Therefore, recognising the most frequent diabetes risk factors is critical to lowering diabetic complications and the economic burden. According to current research, the key risk factors for diabetes are eating habits, age, family history, ethnic groupings, obesity, a sedentary lifestyle, and an earlier diagnosis of gestational diabetes. Previous research has also found that sex, pregnancy, BMI and metabolic condition have a connection to diabetes.

In this report, we will cover the dataset that we got from this website (<https://www.kaggle.com/datasets/samira1992/diabetes-intermediate-dataset/data>), where the publisher's name is Samira Shemirani, and the dataset was last updated in 2024. This data came from the National Institute of Diabetes,

Digestive and Kidney Diseases. The dataset's goal is to diagnose whether or not a patient has diabetes based on various diagnostic metrics such as the patient's number of pregnancies, BMI, insulin level, age, and other factors. Several limitations were imposed on the choice of these instances from a broader database. Every single patient here is a female aged at least 21 [4].

Dataset Description:

Numerical Data	Features	Description
	Pregnancies	This column provides information on the number of times a patient was pregnant.
	Glucose	This column tells about plasma glucose concentration at 2 hours in an oral glucose tolerance test.
	BloodPressure	This column shows diastolic blood pressure (mmHg).
	SkinThickness	This column provides information on triceps skinfold thickness (mm)
	Insulin	This column provides information about 2-hour serum insulin (mu U/ml).
	BMI	This column indicates body mass index (BMI), calculated as weight in kg divided by the square of height in meters.

	DiabetesPedigreeFunction	This column represents the Diabetes Pedigree Function.
	Age	This column provides information about years.
	Outcome	This column in the dataset is the target, where 0 represents 'No' (healthy), and 1 represents 'Yes' (diabetic)

PRE-PROCESSING

Manually:

Step 1: I deleted all the records with a blood pressure of 0 because it is not possible for a living person to have a blood pressure of zero. Its shown on Picture below:

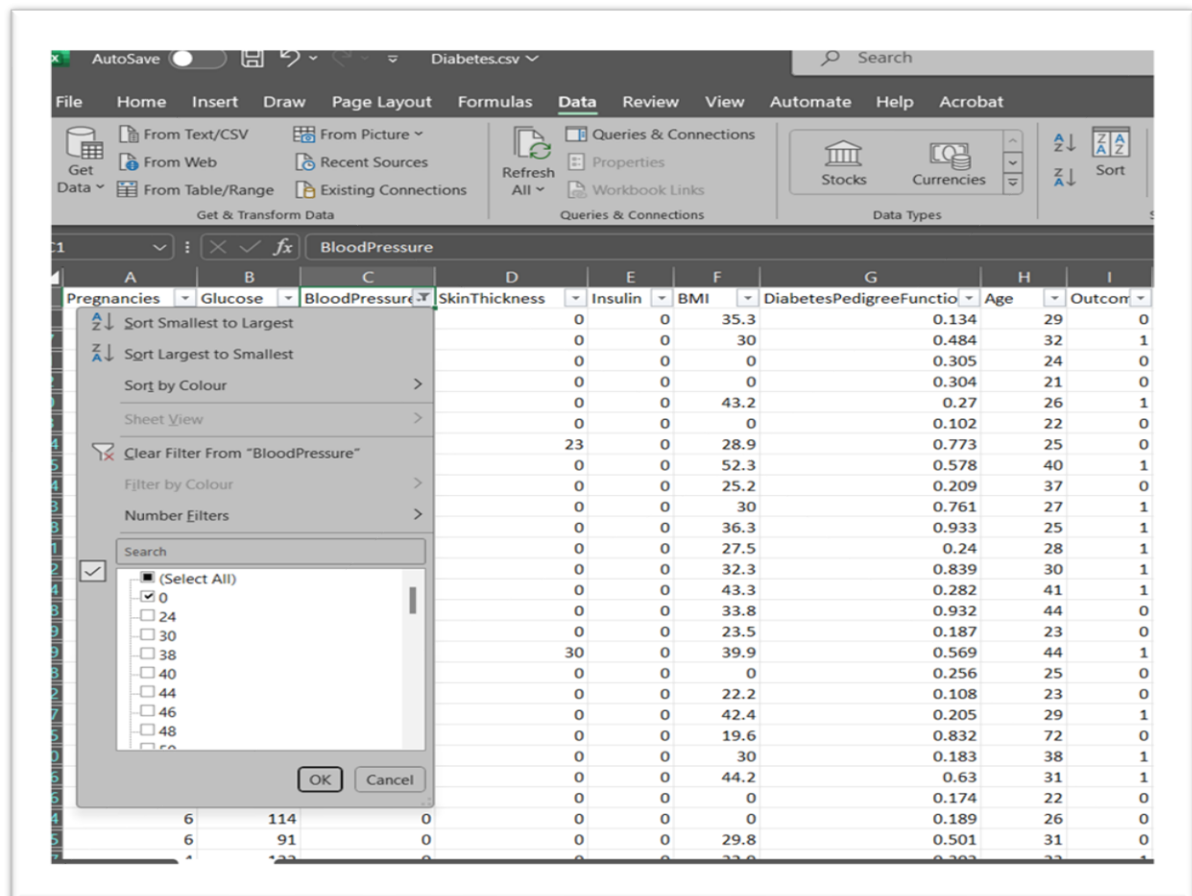


Figure 1

Step 2: I deleted all the records with a BMI of 0 because it is not possible for a living person to have a BMI of zero. The calculation of a person's BMI involves dividing their weight in kilograms by the square of their height in meters. The BMI of an individual would remain higher than zero even if they were severely underweight or had a very low weight as a result of disease, accident, or other extreme circumstances. A person with a BMI of zero would also appear to have no mass at all, which is not possible in the real world.

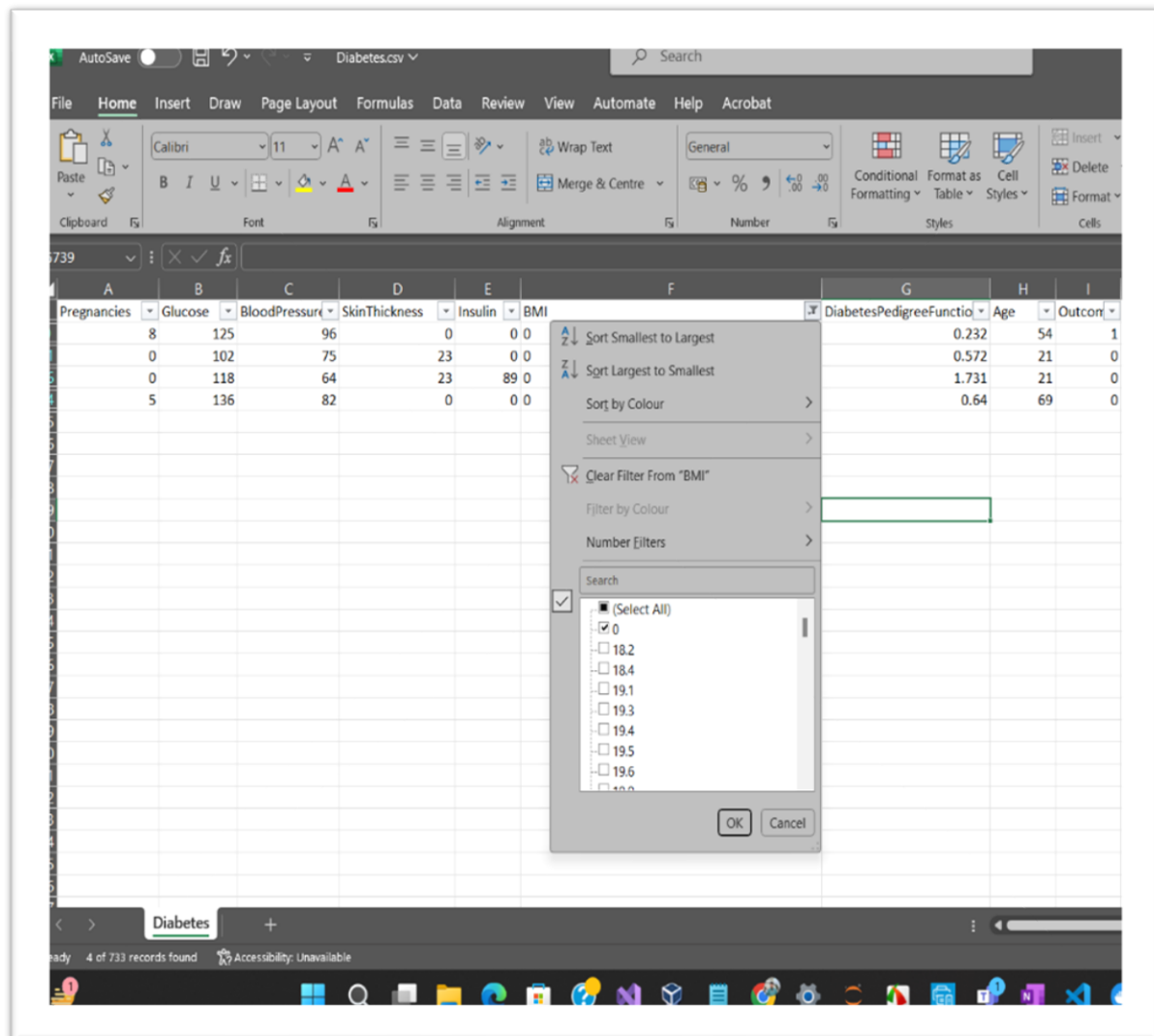


Figure 2

Step 3: The Glucose column has the some invalid values like 0 on few cell. I did remove them.

Shown in picture:

	A	B	C	D	E	F	G	H	I
1	Pregnancies	Glucose	BloodPres	SkinThickn	Insulin	BMI	DiabetesPe	Age	Outcome
77	1	0	48	20	0	24.7	0.14	22	0
184	1	0	74	20	23	27.7	0.299	21	0
344	1	0	68	35	0	32	0.389	22	0
351	5	0	80	32	0	41	0.346	37	1
504	6	0	68	41	0	39	0.727	41	1

Figure 3

Step 4: On SkinThickness column there somany values of 0 which is not possible to someone have a skin thickness of 0 mm. So, using filter on MS Excel remove all the 0 values.

	A	B	C	D	E	F	G	H	I
	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
1									
4	8	183	64	0	0	23.3	0.672	32	1
7	5	116	74	0	0	25.6	0.201	30	0
9	10	115	0	0	0	35.3	0.134	29	0
11	8	125	96	0	0	0	0.232	54	1
12	4	110	92	0	0	37.6	0.191	30	0
13	10	168	74	0	0	38	0.537	34	1
14	10	139	80	0	0	27.1	1.441	57	0
17	7	100	0	0	0	30	0.484	32	1
19	7	107	74	0	0	29.6	0.254	31	1
23	8	99	84	0	0	35.4	0.388	50	0
24	7	196	90	0	0	39.8	0.451	41	1
28	7	147	76	0	0	39.4	0.257	43	1
31	5	117	92	0	0	34.1	0.337	38	0
35	6	92	92	0	0	19.9	0.188	28	0
38	11	138	76	0	0	33.2	0.42	35	0
43	7	133	84	0	0	40.2	0.696	37	0
46	7	159	64	0	0	27.4	0.294	40	0
48	1	146	56	0	0	29.7	0.564	29	0

Figure 4

With Python Command

Step 1: Check for any duplicates in the dataset and remove them:

```
[7] # Check for duplicates and remove them if found
    if diabetes_data.duplicated().any():
        print("Found duplicate rows. Removing them...")
        diabetes_data.drop_duplicates(inplace=True)
```

Figure 5

Step 2: Checking dataset info to find any Null value on it:

```
#Print dataset info
diabetes_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 531 entries, 0 to 530
Data columns (total 13 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Pregnancies                          531 non-null    int64
1   Glucose                              531 non-null    int64
2   BloodPressure                        531 non-null    int64
3   SkinThickness                       531 non-null    int64
4   Insulin                             531 non-null    int64
5   BMI                                 531 non-null    float64
6   DiabetesPedigreeFunction             531 non-null    float64
7   Age                                  531 non-null    int64
8   Outcome                             531 non-null    int64
9   GlucoseGroup                         531 non-null    category
10  BloodPressureGroup                   531 non-null    category
11  BMIGroup                             531 non-null    category
12  PedigreeGroup                       531 non-null    category
dtypes: category(4), float64(2), int64(7)
memory usage: 41.0 KB
```

Figure 6

Step 3: Checking the dataset insights with count, mean, std, minimum value, maximum value etc.

```
[11] #dataset overview with all the mean and std
diabetes_data.describe()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
count	531.000000	531.000000	531.000000	531.000000	531.000000	531.000000	531.000000	531.000000	531.000000
mean	3.514124	121.071563	71.538606	29.184557	115.048964	32.888136	0.502974	31.617702	0.333333
std	3.314532	31.013670	12.298332	10.533676	123.115690	6.887428	0.344871	10.771503	0.471849
min	0.000000	56.000000	24.000000	7.000000	0.000000	18.200000	0.085000	21.000000	0.000000
25%	1.000000	98.500000	64.000000	22.000000	0.000000	27.850000	0.258500	23.000000	0.000000
50%	2.000000	115.000000	72.000000	29.000000	92.000000	32.800000	0.415000	28.000000	0.000000
75%	5.000000	141.500000	80.000000	36.000000	165.500000	36.900000	0.659000	38.000000	1.000000
max	17.000000	199.000000	110.000000	99.000000	846.000000	67.100000	2.420000	81.000000	1.000000

Figure 7

Hypothesis

Hypothesis 1: Identifying Risk Factors of Diabetes Based on Glucose, Blood Pressure, and BMI

A group of doctors and scientists is conducting research to determine the level of glucose, Blood Pressure, and BMI of a patient that puts him at a high risk of Diabetes. They want to know exactly what levels increase the risk of developing diabetes. By determining this, they may alert people early on whether they require medical assistance. This will make diabetes treatment easy and make people healthy.

Hypothesis 2: Age-Based risk of getting Diabetes.

Doctors and scientists are also examining which age groups are most vulnerable to developing diabetes. They want to know which persons in various age groups are most likely to have the disease. Once they understand, they may inform

those age groups about the dangers of diabetes. This way, people in those age groups may take steps to be healthy and seek help when required.

Hypothesis 3: Predicting Diabetes Risk Based on Family History (Diabetes Pedigree Function).

Doctors and scientists are doing important research to predict the risk of getting diabetes based on family history, with a special focus on the Diabetes Pedigree Function (DPF). Their major objective is to determine the efficacy of using DPF along with other health indicators, such as the number of births and skin thickness, in predicting an individual's risk of developing diabetes.

Researchers want to identify those who are at a higher risk of acquiring diabetes or who already have the disease but are unaware of it. By using this knowledge, medical professionals may provide patients with individualised advice on adopting proper dietary habits and lifestyle changes to reduce the risk of diabetes development.

In addition, Early detection is key for those at risk of diabetes due to their genetics. It will allow doctors to act fast, manage the condition, and avoid complications.

Proposed Solution:

The **K-Nearest Neighbour(K-NN)** algorithm can effectively solve the problems proposed in the hypothesis. It is a powerful tool that is capable of handling both classification and regression problems. By finding the nearest neighbour within a certain area around a given point, the algorithm identifies the closest attribute to make an accurate prediction. This technique is commonly known as supervised learning and has proven to be highly effective in various applications.

The diagram below shows the step-by-step process of how the K-Nearest Neighbour works:

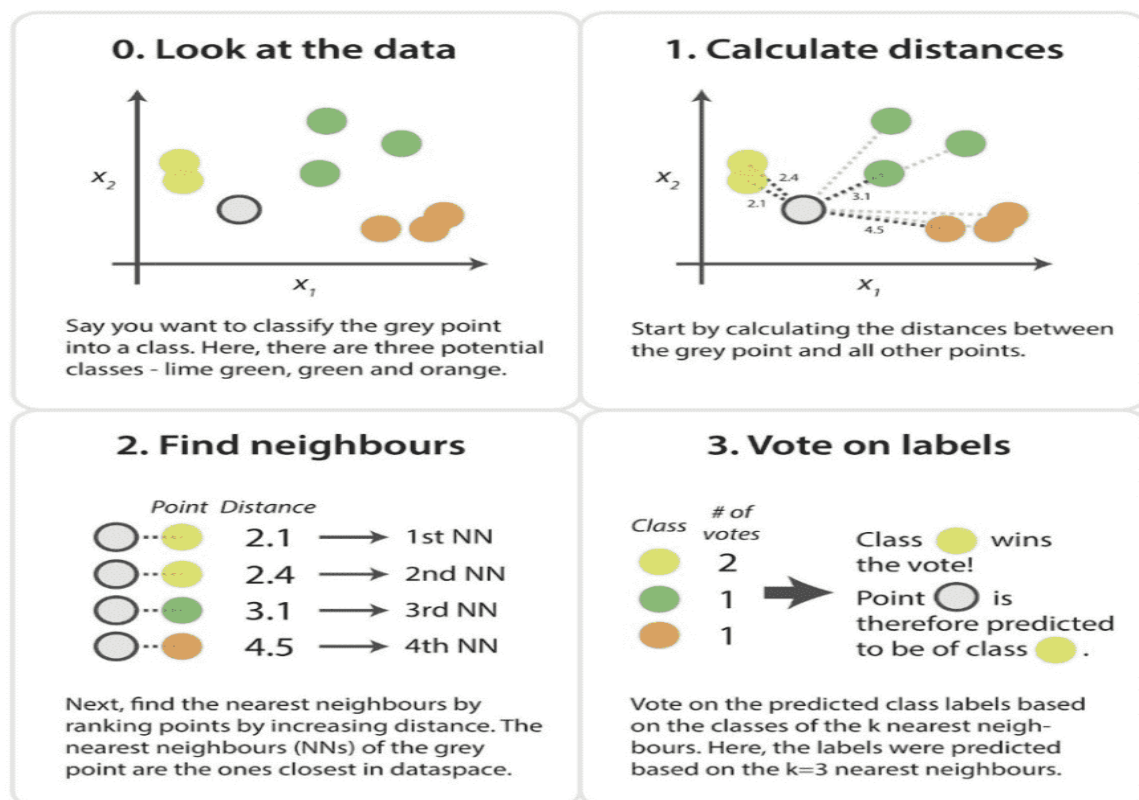


Image: K-Nearest Neighbour © medium.com

Hypothesis 1

For hypothesis 1, the K-Nearest Neighbour method will use patient data such as glucose levels, blood pressure, and BMI measures as predictors to determine whether a person has Diabetes by predicting outcome 1 or 0. Here, outcome 1 means the person has diabetes, and 0 means the person doesn't have diabetes.

For That It will first measure the distance with Neighbour data points. Then, it will find the k number of the smallest distances. After that, it will find the majority outcome, which will be the predicted result.

Step 1: Visualization of the whole dataset with Histogram and Pie Chart to see number of people who have diabetes and who don't and the percentage of it.

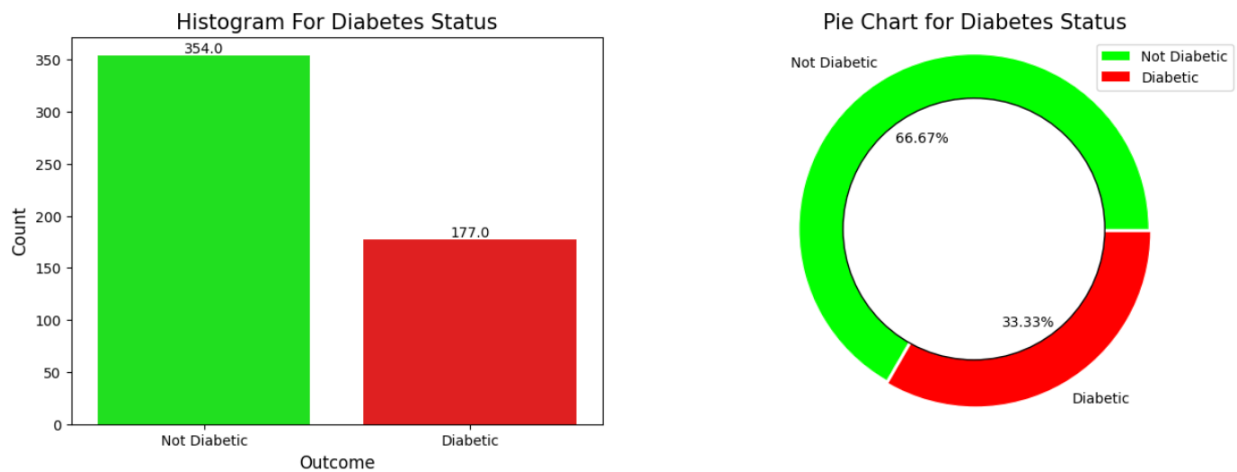


Figure 8: Visualization of the whole dataset with Histogram and Pie Chart

According to the graphs, out of 531 people:

354 people don't have diabetes, almost 66.67%.

177 people have diabetes, almost 33.33%.

Step 2: Training the dataset with the predictors and checking accuracy:

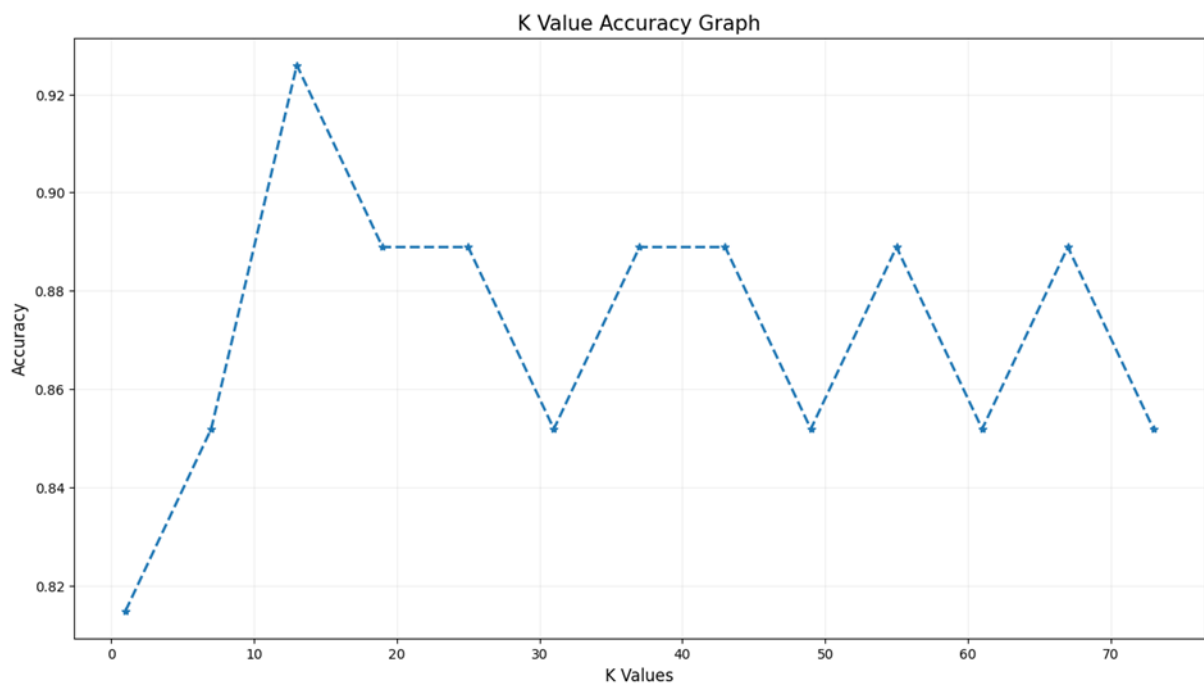


Figure 9: K value accuracy graph.

The best K Value is 13 with 92.59% Accuracy.

Step 3: Visualize and analyse the Glucose, BloodPressure and BMI data against the Outcome with value 1(Have Diabetes):

For each range of Glucose, BloodPressure and BMI, how many people have Diabetes in percentage for each range:

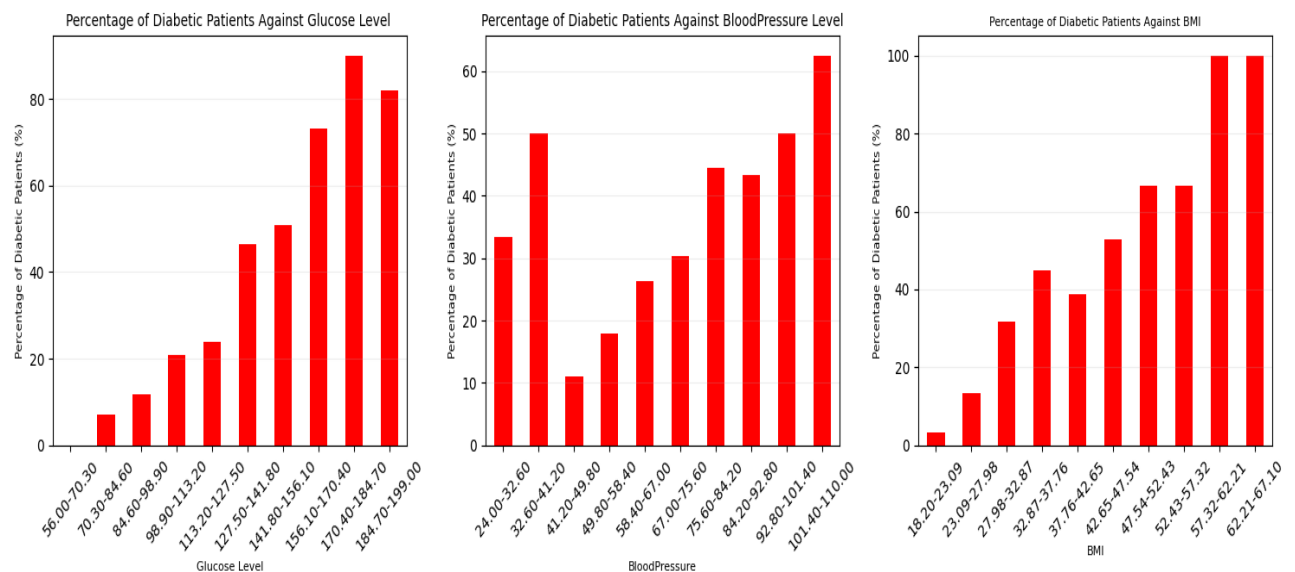


Figure 10: Visualization and analyzation of the Glucose, BloodPressure and BMI data against the Outcome with value 1(Have Diabetes)

From the analysis, it is clear that people who have higher Glucose, Blood Pressure, and BMI are at the highest risk of having Diabetes.

Step 4: Getting raw data from the user and check the prediction:

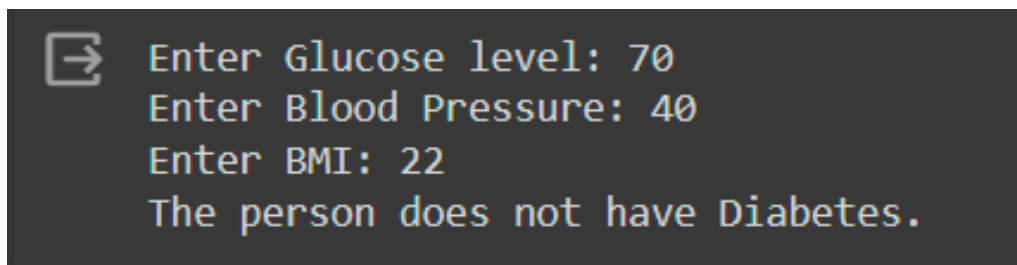


Figure 11: User inputs raw data and gets the prediction accordingly.

Hypothesis 2

For Hypothesis 2, K-Nearest Neighbours (KNN) will use patient data such as age and diabetes status to identify age-related risk variables for diabetes. The algorithm will compute the distance between each patient's age and that of nearby data points. It will then determine the k number of nearest neighbours with comparable ages. KNN will forecast the chance of diabetes development for people of various ages by analysing the most common result among these neighbours.

Step 1: Plotting a histogram to visualize diabetes data according to the age:

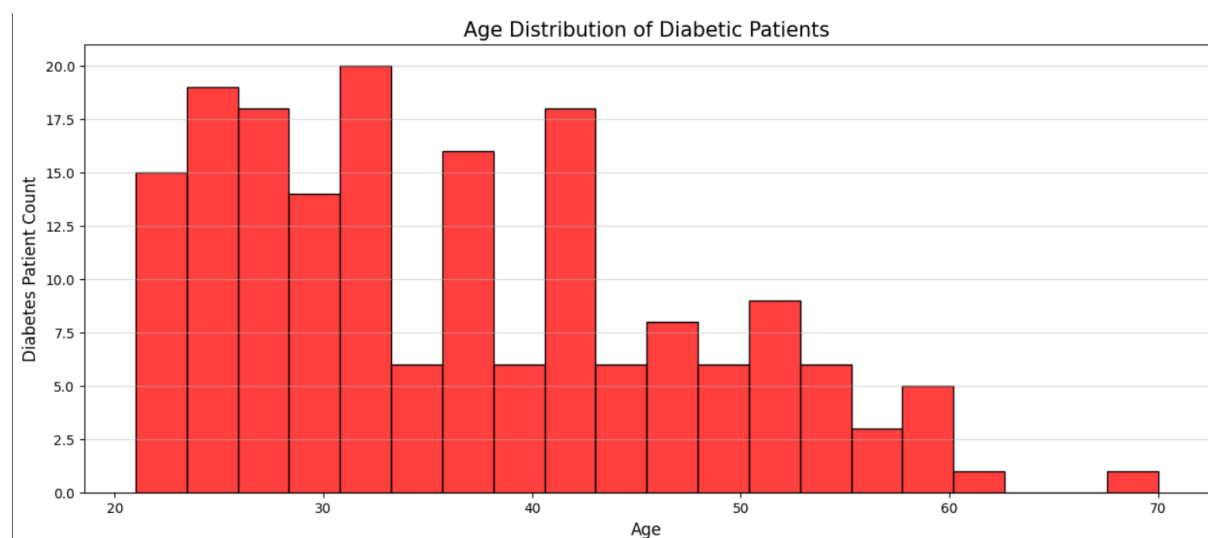


Figure 12: Diabetes by Age

From the Graph above, we can see that most people with diabetes are younger. So, the younger people need to be more careful about the diabetes.

Step 2: Training the dataset with the predictors and checking accuracy:

```
➡ Accuracy: 0.7037037037037037
```

Figure 13

Step 3: Getting raw data from the user and check the prediction:

```
Enter Age: 22
The person is at high risk of Diabetes.
```

Figure 14

Hypothesis 3

For Hypothesis 3, K-Nearest Neighbours (KNN) will use patient data such as Diabetes Pedigree Function (DPF) to predict diabetes risk. The algorithm will determine the distance between each patient's data and that of neighbouring data points, taking into account DPF. It will then discover the k nearest neighbours with comparable characteristics. By analysing the majority result among these neighbours, KNN predicts an individual's likelihood of acquiring diabetes based on their genetic history and other health characteristics.

Step 1: Visualize and analyse the Diabetes Pedigree Function (DPF) data against the Outcome with value 1 (Have Diabetes):

For each range of Diabetes Pedigree Function (DPF) how many people have Diabetes in percentage for each range

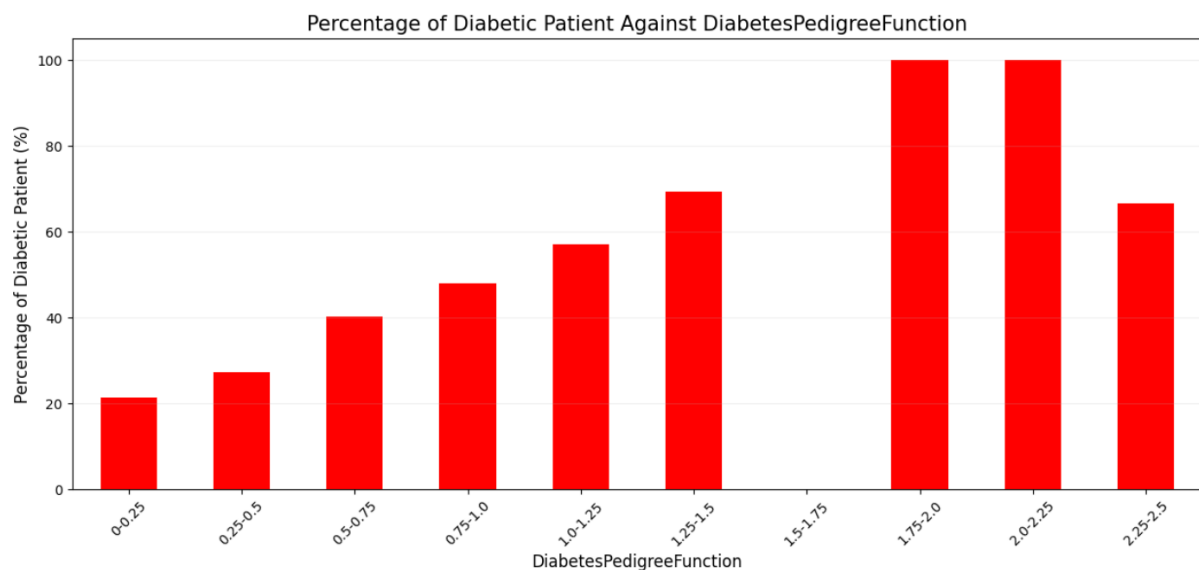


Figure 15

From the Graph, we can see that the people who have higher Diabetes Pedigree Function (DPF) are at a high risk of getting Diabetes.

Step 2: Training the dataset with the predictors and checking accuracy:

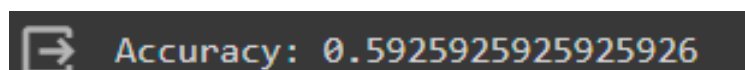


Figure 16

Step 3: Getting raw data from the user and check the prediction:

```
Enter Diabetes Pedigree Function: 2
The person has Diabetes.
```

Figure 17

Reflection:

On coursework 2, I did add a few more pre-processing than coursework 1. In coursework 1, I just did manual pre-processing, but here, I did add some more Python commands for pre-processing, like checking duplicate data, checking null data and removing them from the original dataset.

Reference:

- [1] Diabetes UK "Diabetes facts and stats" 2018. Available: <https://www.diabetes.org.uk/resources-news-expert-opinion/facts-and-stats>
- [2] Colman, P. G., & Donnelly, R. (2017). Diabetes in the UK 2017. Practical Diabetes International, 34(8), 380–382. doi: 10.1002/pdi.2154
- [3] Anonymous "Cost of Diabetes." 2019. Available: <https://www.diabetes.co.uk/cost-of-diabetes.html>
- [4] S. SHEMIRANI, "Diabetes - Intermediate  Dataset," 2022. Available: <https://www.kaggle.com/datasets/samira1992/diabetes-intermediate-dataset>